Name: _____.

Student number: _____.

Instructions

- Turn off all the electronic devices.
- This is a closed book, and no notes exam.
- The last page of this exam contains a notesheet that you can tear off.
- You can use any fact in the notesheet, but you must reference which fact you are using.
- You can use every theorem proved in class, but you need to state the theorem.
- Unless otherwise stated, you must justify your answers.
- If you have a question, raise your hand and I will come to you.
- You have 1 hour and 50 minutes to complete the exam.
- Good luck!

Question:	1	2	3	4	5	6	Total
Points:	10	10	10	10	10	10	60
Score:							

1. (10 points) Let $r \in \mathbb{R}$ with $r \neq 1$. Prove that $\sum_{i=0}^{n} r^i = \frac{1 - r^{n+1}}{1 - r}$ for every integer $n \geqslant 0$.

2. (10 points) Consider the function $f\colon \mathbb{Z} \to \mathbb{Z}$ defined by

$$f(x) = \begin{cases} x + 2 & \text{if 3 divides } x \\ x - 1 & \text{otherwise} \end{cases}$$

(a) Prove that f is injective.

(b) Prove that f is surjective.

3. (10 points) Determine whether the following statements are true or false. If the statement is true, provide a brief proof. If the statement is false, provide a counterexample

   (a) If A and B are sets, then $A \backslash B$ and $B \backslash A$ are disjoint.

   (b) Let $f\colon \mathbb{N} \to \mathbb{N}$ be a function. If f is injective, then it is surjective.

   (c) Let $A = \{n \in \mathbb{Z}\colon n \equiv 3 \pmod 8\}$. If $x \in A$, then x is odd.

4. (10 points) For each $m \in \mathbb{N}$, consider the set $A_m = \{mn \mid n \in \mathbb{N}\}$.

   (a) Is it true that $A_m \approx \mathbb{N}$ for all $m \in \mathbb{N}$? Either prove it or disprove it.

   (b) Find $\bigcup_{m \in \mathbb{N}} A_m$.

   (c) Find $\bigcap_{m \in \mathbb{N}} A_m$.

5. (10 points) Let A and B be nonempty sets and $f\colon A \to B$ a function.

   (a) Let $X \subseteq B$ and $Y \subseteq B$. Prove that $f^{-1}(X \backslash Y) = f^{-1}(X) \backslash f^{-1}(Y)$.

   (b) Let $U \subseteq A$ and $V \subseteq A$. Prove that $f(U \backslash V) \supseteq f(U) \backslash f(V)$.

6. (10 points) Let F be the set of all functions $f\colon \mathbb{R} \to \mathbb{R}$. Define a relation $\sim$ on F by $f \sim g$ if and only if there exists some $c \in \mathbb{R}$ such that $f(x) = g(x) + c$ for all $x \in \mathbb{R}$. Show that $\sim$ is an equivalence relation.

Final Exam Notesheet

Divisibility. For $m, n \in \mathbb{Z}$ with $m \neq 0$, m *divides* n if $n = mq$ for some $q \in \mathbb{Z}$. Denoted $m \mid n$.

(D1) Divisibility is reflexive and transitive.
(D2) If $a \mid b$ then $a \mid bc$.
(D3) The Division Theorem. For $a, b \in \mathbb{Z}$ with $b > 0$, there exists unique $q, r \in \mathbb{Z}$ with $a = bq + r$ and $0 \leqslant r < b$.

Congruence. For $n, a, b \in \mathbb{Z}$ with $n > 1$, we say $a \equiv b \pmod{n}$ if $n \mid (a - b)$.

(C1) If $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$ then $a + c \equiv b + d \pmod{n}$.
(C2) If $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$ then $ac \equiv bd \pmod{n}$.
(C3) If $a \equiv b \pmod{n}$ then $a^m \equiv b^m \pmod{n}$ for any $m \in \mathbb{N}$.
(C4) For any $a \in \mathbb{Z}$ there is a unique $r \in \mathbb{Z}$ with $0 \leqslant r < n$ such that $a \equiv r \pmod{n}$.

Sets. Let A, B and C be subsets of some universal set U.

(S1) $A \subseteq B$ when $x \in A \Rightarrow x \in B$.
(S2) $A = B \Leftrightarrow A \subseteq B$ and $B \subseteq A$.
(S3) $\mathcal{P}(A) = \{D \subseteq U \mid D \subseteq A\}$.
(S4) If $|A| = n$ then $|\mathcal{P}(A)| = 2^n$ for any $n \geqslant 0$.
(S5) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.
(S6) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.
(S7) $(A^c)^c = A$ and $A \backslash B = A \cap B^c$.
(S8) $A \times B = \{(x, y) \mid x \in A \text{ and } y \in B\}$.

Indexed Families of Sets. Let Λ be a not empty (index) set. Suppose that we have a set A_α for each $\alpha \in \Lambda$.

(IFS1) Union: $\bigcup_{\alpha\in\Lambda} A_\alpha = \{x \mid x \in A_\alpha \text{ for some } \alpha \in \Lambda\}$. Intersection: $\bigcap_{\alpha\in\Lambda} A_\alpha = \{x \mid x \in A_\alpha \text{ for every } \alpha \in \Lambda\}$.
(IFS3) For any $\beta \in \Lambda$, we have $A_\beta \subseteq \bigcup_{\alpha\in\Lambda} A_\alpha$ and $\bigcap_{\alpha\in\Lambda} A_\alpha \subseteq A_\beta$.
(IFS4) De Morgan's Laws: $\left(\bigcup_{\alpha\in\Lambda} A_\alpha\right)^c = \bigcap_{\alpha\in\Lambda} A_\alpha^c$ and $\left(\bigcap_{\alpha\in\Lambda} A_\alpha\right)^c = \bigcup_{\alpha\in\Lambda} A_\alpha^c$.

Functions. Let A and B be non-empty sets.

(F1) A function $f\colon A \to B$ is injective if, for any $x, y \in A$, $f(x) = f(y) \Rightarrow x = y$.
(F2) A function $f\colon A \to B$ is surjective if for every b in B there exists some $a \in A$ such that $f(a) = b$.
(F3) A function is bijective if it is injective and surjective.
(F4) If $V \subseteq A$ we denote $f(V) = \{f(x) \mid x \in V\} \subseteq B$. If $Y \subseteq B$, we denote $f^{-1}(Y) = \{x \in A \mid f(x) \in Y\} \subseteq A$.
(F5) If $f\colon A \to B$ is a function we denote $f^{-1} = \{(f(a), a) \in B \times A \mid a \in A\}$.
(F6) If $f\colon A \to B$ is bijective then f^{-1} defines a function $f^{-1}\colon B \to A$ defied by $f^{-1}(b) = a \Leftrightarrow f(a) = b$.
(F7) If $f\colon A \to B$ and $g\colon B \to C$, the composition $g \circ f\colon A \to C$ is defined by $(g \circ f)(x) = g(f(x))$.

Relations. Let A and B be sets. A relation R from A to B is a subset $R \subseteq A \times B$. We denote $a \sim b$ to denote $(a, b) \in R$. A relation on a set A is a relation from A to A.

(R1) A relation $\sim$ on A is an equivalence relation if it is:

- Reflexive: $x \sim x$ for all $x \in A$.
- Symmetric $x \sim y \Rightarrow y \sim x$.
- Transi: ($x \sim y$ and $y \sim z$) $\Rightarrow x \sim z$.

(R2) If $\sim$ is an equivalence relation, the equivalence class of $a \in A$ is $[a] = \{x \in A\colon x \sim a\}$.
(R3) If $\sim$ is an equivalence relation on A, the collection of equivalence classes is a partition of A.
(R4) For $n \in \mathbb{N}$, we denote by $\mathbb{Z}_n$ the set of equivalence classes for the relation of congruence modulo n.

Equinumerousity. Two sets A and B are equinumerous if there is a bijection $A \to B$. Denoted $A \approx B$.

(E1) Equinumerousity is an equivalence relation.
(E2) A is finite if either $A = \varnothing$ or $A \approx \mathbb{N}_k = \{1, 2, \ldots, k\}$ for some $k \in \mathbb{N}$.
(E3) If B is finite and $A \subseteq B$ then A is finite and $\text{card}(A) \leqslant \text{card}(B)$.
(E4) If A is equinumerous to one of its proper subsets, then A is infinite.
(E5) Pigeonhole: If A and B are finite and $\text{card}(A) > \text{card}(B)$, there is no injection $A \to B$.
(E6) Surjective Pigeonhole: If A and B are finite and $\text{card}(A) < \text{card}(B)$, there is no surjection $A \to B$.
(E6) A set A is countably infinite if $A \approx \mathbb{N}$. The sets $\mathbb{Z}$, $\mathbb{N} \times \mathbb{N}$ and $\mathbb{Q}$ are countably infinite.